Bot 1 (pmus) Structural Remediation & Hardening Verification Report

Date: October 26, 2026

Branch: pmus-rebuild

Commits: e3dc7c5, fa2f0ec

Verification Suite: 533 passed in 5.49s (0 failed, 0 deselected)

1. $59.00 USDC Cash Increase Provenance (13:59 CT / 18:59 UTC)

The $59 cash jump on October 26 was not an external bank deposit. It came from three on-chain USDC inflows into proxy wallet 0xacde6b44ed91cb334fbef01d03ef3fc4db0ca8f4:

- 13:59:11 CT **($21.50 USDC)**: Manual UI SELL exit of 59.99 shares of Will Tyreek Hill sign with the Kansas City Chiefs in 2026? across 3 price levels (5.00 @ $0.370, 5.00 @ $0.360, 49.99 @ $0.357).
- 13:59:09 CT **($29.10 USDC)**: Manual UI SELL exit of 55.00 shares of Will no next Israeli Minister of Justice be appointed by February 28, 2027? across 2 price levels (5.00 @ $0.540, 50.00 @ $0.528).
- Combined 13:59 CT liquidation proceeds: $50.60 **USDC**.
- **Earlier Intraday Inflows ($12.05 USDC)**: $2.05 from a 5-share Tyreek Hill sell at 13:33 CT (@ $0.410) plus $10.00 USDC from the daily LIP liquidity reward payout at 00:00:58 UTC.

2. Forensic Gap Review & Residual Risks Addressed

During the deep forensic review, four critical operational hazards were uncovered and structurally resolved:

- **Historical $0.00 Bag Leakage Eliminated**:
- Live query of proxy wallet 0xe7d565d58c61e2b96adb0e4518e7c33978402e7f revealed 16 unredeemed $0.00 positions dating back to October 6 (Albania vs San Marino, Spain, Scotland vs Slovenia, etc.) totaling $38.00+ in sunken losses. Indiscriminately summing these would cause an immediate halt on tick 1 every day.
- _compute_redeemable_loss() was updated to gate losses strictly to positions active or traded *today* (ledger.has_fill_since() since midnight UTC, local_fill_positions, or target_slugs), shielding the bot from ancient dead bag halts.
- **Persistent 24-Hour Toxic Market Cooldowns**:
- loss_cooldown_until was previously stored only in RAM. Process restarts or reboot bounces wiped the table, immediately re-admitting toxic markets.
- Added persistent SQLite storage in loss_cooldowns table. LiveEngine.__init__ now seeds active cooldowns directly from the database on startup.
- **Automated SQLite Corruption Recovery**:
- Ledger.__init__ was hardened to catch sqlite3.DatabaseError (e.g. malformed disk image), auto-backup the corrupted file to ledger.db.corrupt_<timestamp>, clear stale lock files, and cleanly re-initialize the schema without unhandled process termination.
- **Timing-Attack Hardened Dashboard Authentication**:
- Updated dashboard.py password verification to use constant-time hmac.compare_digest.

3. Comprehensive Defect Remediation Summary

- **Post-Fill Synthetic Pair Merge Completion**: `allocate_conditions` and `_waterfill_conditions` enforce `max_leg_collateral`, cap post-fill opposite-leg hedges at `1.00 - filled_avg_price` down to `HEDGE_LEG_MIN_PRICE = $0.05` (with matching `$0.05` floor in `lip_quoter.desired_quotes`), and verify `p_yes + p_no <= $0.99`.
- **Hair-Trigger Stop-Unwind Abolished**: `LIQUID_STOP_UNWIND` requires `allow_stop_unwind=True` and a `>= 15%` drawdown from entry (`mid <= avg_price * 0.85`). Realized loss exits trigger an 86,400s (24h) cooldown on both sibling slugs and zero out active bid budgets immediately.
- **Redeemable $0.00 Loss Recognition & Risk-Stop Ordering**: `LiveEngine.tick()` evaluates `daily_loss_stop` before cash-shift reallocation, incorporates sibling-netted `_compute_redeemable_loss(live_positions)` (excluding `archived` positions and null `cur_price` values) inside `daily_loss_baseline` subtraction, and enforces `PMUS_DAILY_LOSS_BASELINE_DATE`.
- **Taker Multi-Maker VWAP & LIP Scoring Symmetry**: `parse_trade_fills` computes true matched notional across maker legs (including complement pricing `1.00 - maker_price` on cross-asset fills); `side_score` filters sub-`min_size` dust levels; `estimate_global_pair_score` strips own resting orders before computing `adjusted_mid`.
- **SQLite WAL Pruning, Webhook Concurrency & Fail-Closed Dashboard**: `Ledger` caches schema columns at init and prunes snapshots older than 7 days; `Notifier` bounds concurrency with `BoundedSemaphore(4)`, redacts webhook URLs/tokens in error logs, and reports `engine_state` and `mtm_equity`; `dashboard.check_password()` fails closed unless `PMUS_DASHBOARD_PASSWORD` or `PMUS_DASHBOARD_ALLOW_UNAUTH_LOCAL=1` is set.

4. Verification Receipt

`text

& "d:\neststock\scripts\polymarket_bot\venv\Scripts\python.exe" -m pytest pmus/tests -q

533 passed in 5.49s

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